Physica Scripta

RECEIVED
1 October 2024

REVISED
12 November 2024

ACCEPTED FOR PUBLICATION
29 November 2024

PUBLISHED
11 December 2024

PAPER

Galvanostatic performance of single-particle Li-ion battery materials: a rapid diagram diagnosis assisted by a Python / C++ software

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Keywords: lithium-ion batteries, single-particle, materials diagnosis, software

Supplementary material for this article is available online

Abstract

Herein, a novel open-source software for predicting the performance of single particles of electrode materials under galvanostatic charging conditions is presented. The model improves previous work by incorporating different thermodynamic approaches to describe the interaction between intercalated ions, and the software provides tools for fitting different kinetic parameters and generating potential/capacity profiles or concentration/distance profiles to be analyzed. These features allow for detailed studies of the factors limiting the charging rate of active materials in which ions are intercalated. The diagnostic diagram is constructed through simulations of the capacity reached for a given potential cut-off, represented in the domain of two dimensionless parameters. These parameters represent kinetic and particle-size limitations for Li-ion storage. The present tool aims to facilitate the analysis of single-particle experiments, taking advantage of two different computational languages, popular in scientific computing: Python and C++. The software integrates a Python interface, for user-friendly interaction, with a C++ computational core allowing parallelization via OpenMP, for high computational performance. The software supports different thermodynamic approaches: Langmuir-Frumkin intercalation model and equilibrium potentials derived from experimental data. The inclusion of experimental insertion isotherms allows for the construction of realistic capacity diagrams. The open-source software is available at https://github.com/fernandezfran/galpynostatic/.

1. Introduction

Lithium-ion batteries (LIBs) have led to a change in the energy-storage paradigm. Their commercial version (electrodes composed of graphite and $LiCoO_2$ or similar) is based on the intercalation of ions inside the electrodes. Something similar is expected to occur with other technologies based on alkaline ions. However, the urgency in developing safe devices [1, 2] using better-performing active materials, with minimum charging times and maximum capacities, is crucial for the batteries demanded by the industry. Following this direction, extremely fast-charging batteries are needed [3–6].

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To understand the relevance of different factors determining the charging rate and maximum capacity achieved by a material, an exhaustive and systematic study of the materials must be carried out. This process would involve hundreds or thousands of experiments, something impractical. To overcome these difficulties, a computational strategy should allow us to fine-tune the key parameters characterizing this system. Equivalent circuit models are frequently employed for this purpose because of their simplicity and efficiency. However, these models are purely phenomenological, i.e. their parameters are adjusted to replicate empirical data and the internal variables have no physical meaning. Conversely, the most widely used physics-based electrochemical model in the literature, known as the pseudo-two-dimensional (P2D) model, is computationally intensive [7–12].

In recent years, different mathematical approaches have been proposed to standardize the comparison between LIBs materials. The work of Xia *et al* [13] proposes the construction of a figure of merit, say a time Tau given by: $\tau = d^2/D$, where d is the size of electrode particles and D is the diffusion coefficient of ions inside the host. These considerations assume extremely fast kinetics at the electrode/electrolyte interface and semi-infinite diffusion. Following this argument, decreasing particle size or increasing the diffusion coefficient would lead to fast charging. However, the authors highlighted the need to incorporate other important features to improve the model, like the interfacial kinetics.

In an alternative approach, the work of Mustarelli *et al* has proposed different representations to analyse material performance using density functional theory and mathematical models. This approach proposes standardisation of parameters and procedures for the normalisation of a wide range of intercalation cathode materials. The result is a set of simple equations relating the inverse of the state of charge to the square root of the C-rate [14–16]. However, it is unclear how this procedure should be used to model fast charging.

At the single particle (SP) level, three dynamic factors determine the charging rate: the diffusion of ions in the solid, the charge transfer at the electrode/electrolyte interface and the mass transport in the solution [17]. The least limiting of the three is the electrolyte dynamics, which becomes practically irrelevant in single-particle measurements, even at C rates as high as 1000 C for LCO [18].

Following the goal of modelling fast charging at the SP level, a theoretical framework with SP simulations has recently been developed to construct diagrams for materials diagnosis [19]. In this previous work, we studied the performance of materials using an 'ideal' model, where a Nernst approach was implemented to relate the equilibrium potential with the lattice occupation. This means that such an approach ignores the energy change of the inserted ions with lattice occupation and neglects the interactions between the intercalated ions. This model was valuable for general purposes like comparison between different materials [20]. The diagram constructed with this model, predicts the maximum capacity that can be reached by a material under galvanostatic conditions as a function of four quantities: particle size, diffusion coefficient, charge transfer rate constant and C-rate. This information is represented handily since the diagram reduces the four-dimensional problem to a two-dimensional one with two dimensionless parameters: a size/diffusion parameter (ℓ) and a kinetic parameter (Ξ). This allows different predictions, such as the optimal rate to achieve fast charging of the entire active material, as well as comparisons between the performance of different materials, figure 1. The figure schematically illustrates the use of the model diagram to optimize the capacity of a material that can be obtained in practice, by changing some of its characteristics. As seen, the diagram shows how the optimal performance of a material can be achieved (top yellow region of the diagram), by switching values of particle size, diffusion coefficient or kinetic rate constant along the dotted lines.

However, the previous model has some limitations, since it applies only to a general study of materials, as it seeks to make comparisons between them. To conduct more specific studies, sophisticated models are needed in which other factors are incorporated. The particular thermodynamics of each intercalation material, manifested in the equilibrium intercalation isotherm, is one of these factors. Thus, the galvanostatic diagrams must be modified to address particular characteristics depending on the applied thermodynamics. The main purpose of the present article is to analyze the influence of thermodynamics on galvanostatic diagrams and describe the open-source software developed by the team for this aim.

Python and C++ are two of the most popular scientific programming languages. While Python offers the advantages of easy use and data processing, C++, while more difficult to learn, has greater computational power [21]. For this reason, in this work, we present software to build capacity diagrams in the log(Ξ)-log(ℓ) domain, consisting of a Python interface for the user and a C++ core for computational calculations. Furthermore, the present model exhibits a novel feature: it extends the previous one [19], by considering the intercalation isotherms to design the capacity diagrams, making it system-specific and thus more accurate.

The work is organized as follows: firstly, in the 'Electrochemical Model' section, the mathematical model used for the numerical simulations is detailed. In the section 'Maximum State of Charge Diagram Under Galvanostatic conditions' we comment on the theoretical foundations of these diagrams and how to construct them, as well as a practical example of use. In the section 'Software Workflow' we detail the algorithmic design for the software development and in 'Software Benchmarking' we evaluate the software performance. Examples

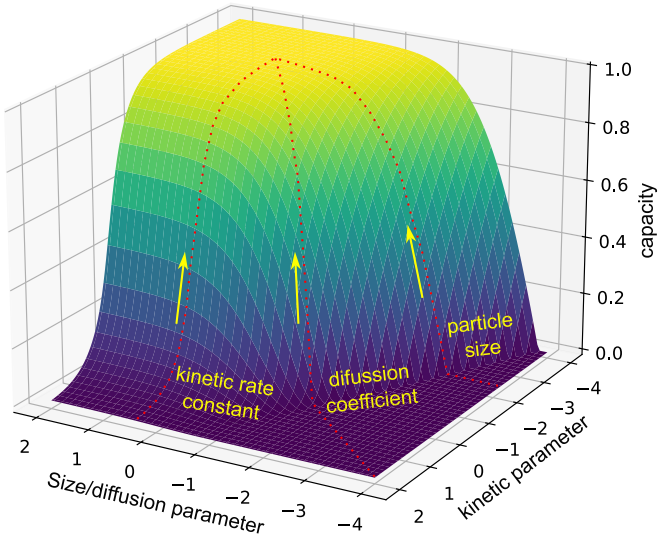

Figure 1. Illustration showing the maximum capacity that can be reached by a single-particle material under galvanostatic conditions, over a two-dimensional domain representing size/diffusion/kinetic limitations. The dotted lines show paths to the high-capacity region from the low-capacity one, obtained by changing the particle size, diffusion coefficient or kinetic rate constant.

of constructed diagrams are given in 'Results' and final comments and perspectives regarding to future software developments, especially with the use of machine learning of increasing relevance in material design [22], is provided in 'Conclusions and Perspectives'.

2. Electrochemical model

As commented previously, the present software aims to construct maps to diagnose LIBs active materials. For such purpose, hundreds or thousands of numerical simulations are needed, since each point in the diagram is obtained from one of these simulations.

To simulate Li^+ intercalation processes, a suitable mathematical model must be established. This model must account for the most relevant features of the chemical environment under investigation. A typical LIB cell consists of composite electrodes made of active materials, conductive additives, a binder and a current collector foil. Therefore, it is laborious to extract intrinsic kinetic information about active materials to understand the rate performance limiting factors. A step in this direction, i.e., to study the electrochemistry of pure active materials, was taken a few years ago with single-particle measurements (SPM) [23–26]. In a conventional SPM, the SP of active material immersed in an electrolyte solution is connected to a metal filament that acts as the current collector. The counter electrode is a Li metal sheet with a large area relative to the active particle.

This scenario can be properly replicated with SP models [18], as well as others [27, 28]. A schematic representation of the SP model used in the present work is shown in figure 2(a): the SP is embedded in an electrolyte surrounded by a Li metal foil as a counter electrode large enough to avoid diffusional limitations. The assumptions for this model are:

- A constant current density i_c (introduced in terms of the C-rate, C_r) in the model, is set to simulate a galvanostatic framework.
- Charge transport inside the particle is primarily limited by the motion of inserted cations, assuming fast electronic transport.
- Ion diffusion within the host follows 1D Fick's second law for a planar ($z = 1$), cylindrical ($z = 2$), or spherical ($z = 3$) geometry (equation (1)). In this equation, t [s] is time, x is the fraction of lithium occupation, r [cm] is the distance from the centre of the particle to the electrode surface, d is the diffusion length and D [cm^2s^{-1}] is diffusion coefficient. The occurrence of different phases is not accounted for in the model.
- The diffusion coefficient of the ions intercalated in the solid (D) remains constant throughout the intercalation process.
- Charge transfer at the electrode/electrolyte interface is governed by the Butler-Volmer approach (equations (3) and 2). In these equations, ρ [$g.cm^{-3}$] is the phase density of the material, Mr [$g.mol^{-1}$] is the

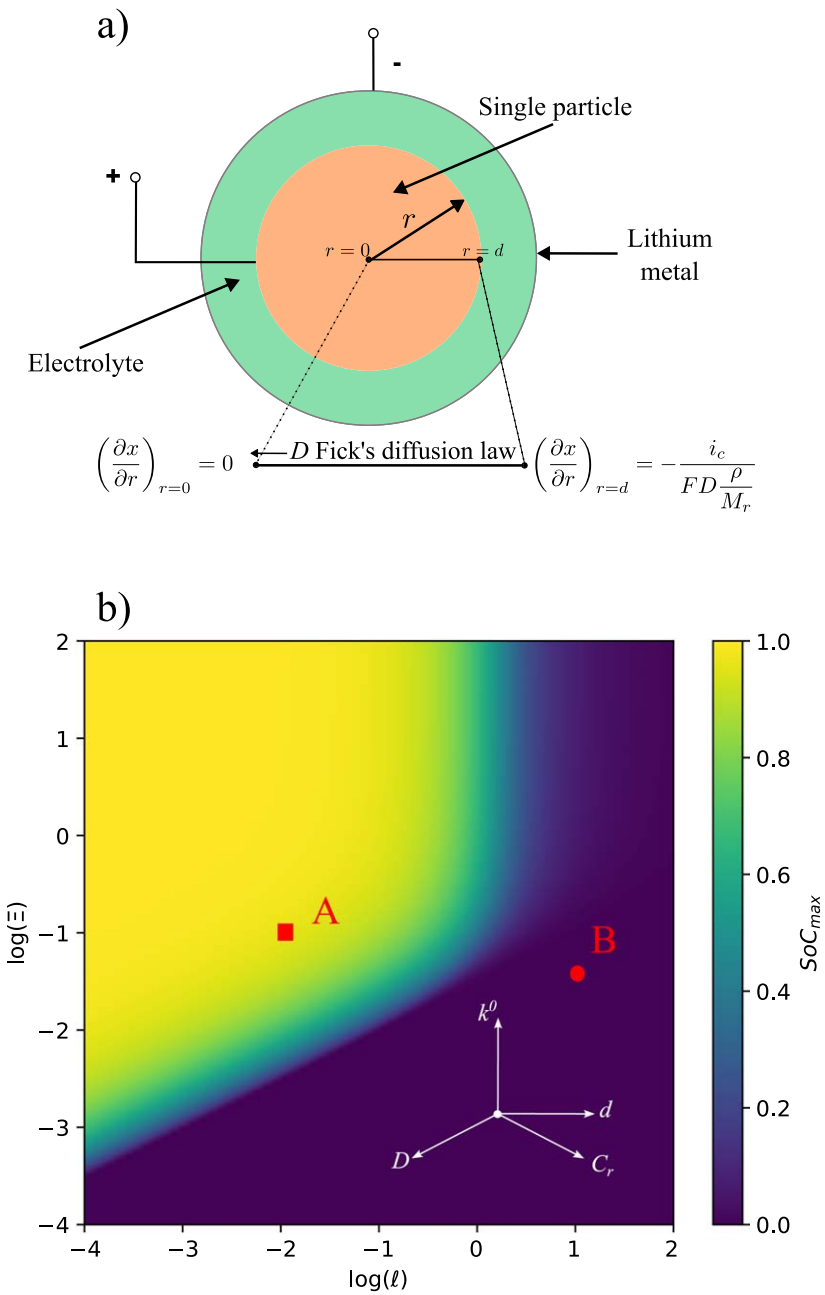

Figure 2. (a) Scheme of a single-particle electrode cell. (b) Schematic coloured diagram obtained from galvanostatic simulations for a spherical single-particle host material without considering Li-Li interactions. $E_{cut} = -150$mV. The arrows show the directions of trajectories on the diagram that would be obtained by increasing one of the parameters, k^0, D, d or C_r, starting from the white dot and taking the remaining parameters as constant.

molecular mass, k^0 [cm.s^{-1}] is the heterogeneous rate constant and R_Ω [Ω.cm^{-2}] is the resistance of the cell. F, R and T have their usual meaning.

- The equilibrium potential, $E_0(x_d)$ [V] is considered to depend on the surface concentration of ions x_d.
- The heterogeneous rate constant (k^0) is assumed to be the same over all the intercalation processes.
- von Neumann boundary conditions were assumed at the centre of the particle, $r = 0$ (equation (4)), and at the surface, $r = d$ (equation (5)).
- A high concentration of ions in the electrolyte and a large diffusion coefficient of these species there allow the neglection of mass transfer within it.
- Volume changes of particles during loading/unloading are not considered.

- Migration under an electric field and side reactions are ignored.

$$\frac{\partial x}{\partial t} = D\left[\left(\frac{\partial^2 x}{\partial r^2}\right) + \frac{(z-1)}{r}\left(\frac{\partial x}{\partial r}\right)\right] \tag{1}$$

$$i^0 = Fk^0\left(\frac{\rho}{M_r}\right)\sqrt{x_d(1-x_d)} \tag{2}$$

$$E = E^0(x_d) + \frac{2RT}{F}sinh^{-1}\left(\frac{i_c}{2i^0}\right) \tag{3}$$

$$\left(\frac{\partial x}{\partial r}\right)_{r=0} = 0 \tag{4}$$

$$\left(\frac{\partial x}{\partial r}\right)_{r=d} = -\frac{i_c}{FD\frac{\rho}{M_r}} \tag{5}$$

Diffusion equations are solved with the implicit finite differences method called the Crank-Nicholson method [29] using the Thomas algorithm. With this purpose, a space-time grid consisting of *time_steps* time points and *grid_size* space points was implemented.

In summary, within the present model, the capacity achieved by a material under galvanostatic conditions depends on four parameters: k^0, D, d and C_r, and the particle geometry. Simulations stop when the electrode potential E reaches a pre-defined cut-off potential, E_{cut}.

In [19], we discussed the impact of the limitations of the model on predicted capacities, but we want to take a few words to elaborate on this.

The most critical point of the model is to neglect the dynamics of the electrolyte. Thus, the present model is valid for C-rates where the concentration of the electrolyte at the particle/solution interface does not vary appreciably relative to that of the bulk concentration of the liquid phase. In this regard, we have developed a simple scheme to estimate those C-rates where the lithium concentration in the electrolyte varies relative to the bulk concentration, this is shown in Supplementary information. These calculations estimate the C-rate at which the Li^+ surface concentration will differ appreciably from the bulk concentration by some percentage. Section S2 of the Supplementary information shows the mathematical development of the limiting C-rates and its impact on the accuracy of the model on the final SoC.

Also, the predicted State-of-Charge will deviate from the experimental value if particle volume changes upon loading, if electronic transport is limiting the lithiation process, if side reactions occur during loading or if the cell's internal resistance is not negligible. Cell internal resistance can be easily introduced in the present model, this can be done by introducing the iR term in the overpotential term of the Butler-Volmer equation. However, since this is a parameter that depends on the experimental arrangement, we recommend correcting the experimental potential/capacity curves to compare with the present modelling. In the conclusion and perspectives section, we propose future improvements to the model.

In [20] we validated the model with experimental results, showing that it can be useful to extract diffusion coefficients D and rate constants k^0 from experimental data, using an heuristic approach. The Ds and k^0s extracted from the model fitting of experimental data were in good agreement with values of these parameters found in the literature [20]. Future modelling may consider some factors omitted here to refine the approach. These modifications will be implemented in further versions of the software.

The equilibrium potential $E_0(x_d)$ is determined by the thermodynamics of the system. Thus, the present approach allows switching $E_0(x_d)$ between two different models: 1) a Langmuir/Frumkin intercalation isotherm (section 2.1) and 2) a cubic spline interpolation of the experimental data points (section 2.2). The Frumkin intercalation isotherm is very useful in illustrating how effective attractive/repulsive interactions contribute to shaping the SoC $_{max}$ maps.

This implementation is a novelty of the present work to construct the diagrams and the impact on it. The two thermodynamic approaches are described in detail in the following sections.

2.1. Langmuir/Frumkin isotherm and concentration profile

The Frumkin intercalation model to describe the thermodynamics of LIBs materials is widespread [30]. The equilibrium isotherm depends on an interaction parameter denoted with g (equation (6)). This parameter can take different values, determining the interaction type: $g = 0$ (non-interacting ions, Langmuirian system), $g > 0$ repulsive interactions, $g < 0$ attractive interactions.

$$E^0(x_d) = E^0 + \frac{RT}{F}\left[g(0.5 - x_d) + ln\left(\frac{1 - x_d}{x_d}\right)\right] \tag{6}$$

In equation (6), E^0 is a term independent of the surface lithium concentration x_d.

This model has been used to characterize different host materials, like LCO ($g = -4.2$) [31], stage II-I transition in graphite ($g = -4$) [32], and $Li_xV_2O_5$ ($g = -3.9$) [33]. Also, it was recently revisited to study the effect of interactions and geometry on the voltammetric behaviour of LIBs materials [34–36].

2.2. Intercalation isotherms constructed from interpolation of experimental data

In this approach, the equilibrium potential $E^0(x_d)$ was calculated using N_{iso} experimental data points, obtained from galvanostatic experimental measurements at a very slow charging rate.

To calculate $E^0(x_d)$, a cubic spline interpolation, equation (7), was implemented:

$$E^0(x_d) = a_j + b_j(x_d - x_j) + c_j(x_d - x_j)^2 + d_j(x_d - x_j)^3 \tag{7}$$

In this equation a_j, b_j, c_j and d_j are cubic spline coefficients and x_j is the experimental lithium ion occupation for one of the points of the equilibrium curve, with $j = 0, 1, \ldots, N_{iso} - 1$. Once x_d is obtained, the code finds the segment of the cubic spline, $x_i \leqslant x_d < x_{i+1}$, to which it belongs. Once the section is found, equation (7) is applied to calculate $E^0(x_d)$. The a_j, b_j, c_j and d_j coefficients are computed with the `CubicSpline` function of the `scipy.interpolate` package [37].

3. Maximum state of charge diagram under galvanostatic conditions

A schematic representation of the galvanostatic diagram was shown in figure 1, to understand the conceptualization of the theory before delving into the equations. In this section we will go further, to taste a flavour of the theory behind the concept.

An electrode's state of charge (SoC) is the ratio between the capacity reached at a given time and the maximum available capacity of a material. Therefore, we define here the term maximum state of charge, SoC_{max}, to refer to the SoC that is reached at a given cutoff potential (E_{cut}). This cutoff potential is determined for practical reasons: in the case of lithiation of anode material, it must be high enough to avoid lithium plating of the material. In the case of delithiation of a cathode material, this must be low enough to avoid complete delithiation of this material, which can be detrimental to its stability.

A galvanostatic diagram [19], figure 2(b), is thus defined as a log-log plot that shows the SoC_{max} that a material is capable of hosting, given four parameters that were defined in the previous section: charge transfer rate constant (k^0), diffusion of ions in the material (D), particle size (d) and C-rate (C_r), for a E_{cut} and particle geometry. Two dimensionless parameters are employed to construct these plots, Ξ and ℓ, as given in equations (8) and (9). In these equations, t_h is the time for one hour in suitable units.

$$\Xi = k^0\left(\frac{t_h}{C_r D}\right)^{1/2}, \tag{8}$$

$$\ell = \frac{d^2 C_r}{z t_h D}. \tag{9}$$

These definitions allow reducing a four-dimensional space (k^0, D, d and C_r) into a two-dimensional one (Ξ and ℓ); making it an easier problem to deal with. It also reduces a family of solutions to a single point on the diagram. All the combinations of k^0, D, d and C_r leading to the same potential profile (and so the same SoC_{max}) yield the same point there. The complete derivation of Ξ and ℓ is provided in the Supplementary information for a planar geometry.

The plot is constructed with the software by defining several Ξ and ℓ points (N_Ξ and N_ℓ, respectively) and saving the data of the maximum capacity, SoC_{max}, reached at a defined cut-off potential (E_{cut}) from the simulation of galvanostatic transients at each of the points. In the present work we considered a number of points $N^2 = N_\Xi \times N_\ell$ with $N_\Xi = N_\ell$. In principle, this construction depends on the mathematical model used to describe the energetics of the Li ions along the intercalation process, which is described by the intercalation isotherm. Therefore, depending on the intercalation model, different diagrams can be constructed with different purposes, as we will show later. As a practical guide, an example of the use of this diagram is given below.

Despite the peculiarities of each system, a diagram that assumes a Langmuirian insertion isotherm ($g = 0$ in equation (6)), is particularly interesting, since, as in the case of the ideal gas model, it describes a universal problem. A universal diagram is thus highly relevant, as it is useful to compare the performance of different materials and make comparative assessments. In recent work, we proposed a universal map with this aim, taking a Langmuirian approach with the mathematical model detailed in the previous section [20]. Figure 2(b)

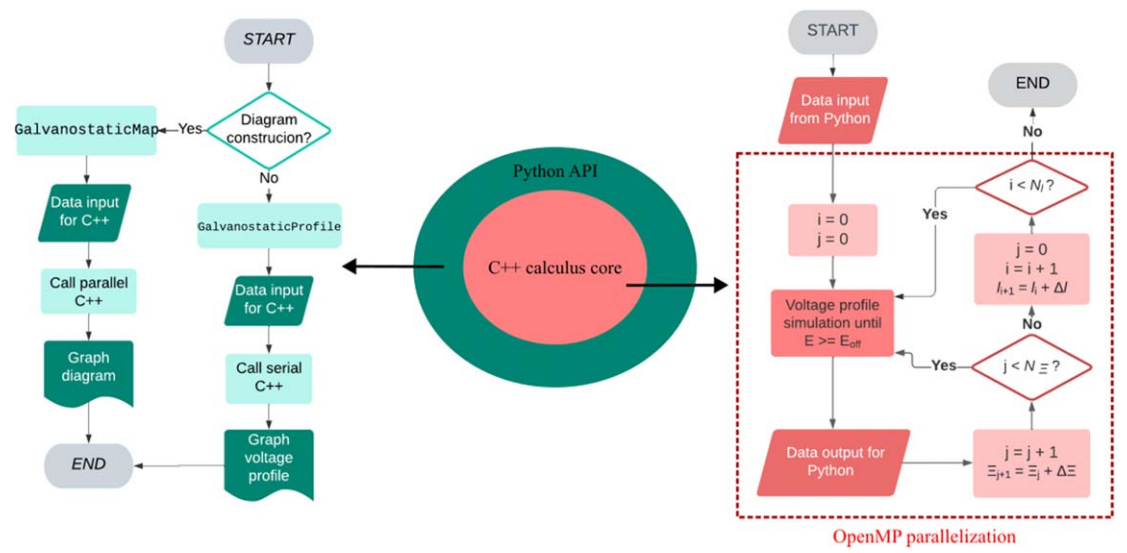

Figure 3. Package workflow for voltage profiles and diagrams simulations.

Table 1. Set of values considered for two representative cases in the diagram of figure 1.

Material	D[cm^2/s]	k^0 [cm/s]	d[cm]	C_r	log(ℓ)	log(Ξ)
A	9.4×10^{-9}	2.03×10^{-7}	0.0005	4	−2.0	−1.0
B	9.2×10^{-12}	3.2×10^{-9}	0.0005	4	1.0	−1.5

compares two different materials in such a universal approach, say A and B, considered as representative cases for systems without interactions between Li ions. The material called 'A' is marked in the universal diagram with a square while the material 'B' is marked with a circle. The set of parameters used for each of them is detailed in table 1. As can be seen, particular combinations of these parameters yield different values of Ξ and ℓ (calculated with equations (8) and (9)) giving different SoC $_{\max}$ and placing the materials in different regions of the map. The map clearly shows that, although the C_r is the same in both cases, the maximum capacity achieved by material A is higher than that of material B.

4. Software workflow

The present extension to the software was designed to take advantage of the computational power of C++ and the readability of Python. It consists of a C++ computational core, where simulations are performed, wrapped into a Python API for easy usage (figure 3). The workflow employs the Object Oriented Paradigm [38], in which the code is organized in classes and objects that are instances of the classes. The Python package `galpynostatic` is available on GitHub and PyPI, https://github.com/fernandezfran/galpynostatic) [20]. The new computational modulus, used in the present work, is called `simulation`. Once the modulus is imported, one can instance a class to simulate a galvanostatic map with `GalvanostaticMap` or a chronopotentiometry profile with `GalvanostaticProfile`. To select one of these classes it is only necessary to configure some input parameters. For `GalvanostaticMap`: E_{cut} must be modified if an equilibrium isotherm is used. For `GalvanostaticProfile`: log(Ξ), log(ℓ) and the equilibrium isotherm. The rest of the parameters are set by default and can be easily changed if necessary.

After choosing any of the main classes, the following step is to execute the `run()` method. When the user executes this method the python API feeds all the parameters into the C++ calculus core through `ctypes`. In the case of the `GalvanostaticMap` class, the C++ code runs the N^2 points of Ξ and ℓ in a parallel mode using OpenMP [39]. The user can set the number of threads. By default, the software uses all the available CPU threads. In the other case, `GalvanostaticProfile` class, C++ run a serial simulation for a given Ξ and ℓ point.

The section of the C++ code that was parallelized is marked in figure 3 inside the red dashed square. The clause `pragma omp for collapse (2)` has been implemented to transform two nested loops into a single

loop whose size is the product of the number of iterations of each loop. The size of the 'collapsed' loop is N^2, since the code runs over the nested loops with sizes N_Ξ and N_ℓ. OpenMP distributes all possible combinations of (ℓ, Ξ) points across threads. Increasing N^2 improves the quality of the plot.

Finally, when the simulations are finished, C++ returns the output data to Python, where it is possible to plot the results using `map_plot` for a galvanostatic map and `isotherm_plot` for a voltage profile. Dataframes are available if the user wishes to save the results in a numerical format using `map_dataframe` and `profile_dataframe` respectively. It is also possible to interact with other existing modules of the software, which is explained in another paper [20].

Another two classes that are not shown in the workflow were also implemented.

The `GalvanostaticProfile` class generates and plots a concentration profile for a given SoC value. This is done using the methods `concentration_dataframe` and `concentration_plot` respectively.

The `ProfileFitting` class fits a potential out-of-equilibrium profile by exploring the space of ℓ and Ξ. It uses the `curve_fit` library from the `scipy.optimize` package. Once the best fit has been found, the equations (8) and (9) can be used to obtain the parameters of interest. For example, given the experimental C_r and particle size d (parameters that are usually controlled experimentally), these equations can be used to calculate k^0 and D.

4.1. Software benchmarking

This section analyses the performance of the software. The software was tested on an Intel(R) Xeon(R) CPU E5-2680 v4 running at 2.40 GHz with two threads per core, fourteen cores per socket and two sockets. The computer had 126 GB of DDR4 memory at a speed of 2667 MT/s with the g++ 13.2.0 compiler. Simulations were run using the GNU compiler with the $-O3$ and $-march = native$ flags. Previous test simulations have shown that these are the best compilation options to optimise the code on this machine and with this compiler (results not shown). These compilation options are defined by default in the software installation via PyPI. Calculations were made with different numbers of points in the time grid, *time_steps*, and fixed space grid, *grid_size*. We also varied the number of CPU threads and the number of diagram points, N^2.

First, the optimal number of *time_steps* was studied for a constant *grid_size* = 1000. A Langmuir isotherm and a planar geometry were used. A Mean Average Error (MAE) was calculated by comparing the simulated data set with a plot calculated using the integral solutions shown in section S1 of the SI. This was done by calculating the mean value of the difference between the SoC_{max} of both calculation methods for $N^2 = 64 \times 64$ points. Figure 4(a) shows the MAE variation with *time_steps*. It can be seen that at *time_steps* = 100000 the MAE reaches a practically constant value. According to the heuristic method of the elbow rule, this point is sufficient to perform the calculations due to the time saving and the gain in accuracy. Therefore *time_steps* = 100000 was chosen as the default value for the software.

To test the simulation time performance, the default *time_steps* was compared with *time_steps* = 3000000. Figure 4 (b) shows the timeruns varying the number of threads and N^2. As expected, in both cases the timerun is larger as N^2 increases and fewer threads are used. However, there is a significant time saving when using *time_steps* = 100000, with a negligible increase in numerical error. According to this analyses, to give realistic numbers, the software takes approximately 13 minutes to perform $128 \times 128 = 16384$ independent simulations using 16 threads, using the default number of *time_steps*. However, to construct a diagram with good resolution, fewer points are required.

A better way to understand timerun performance is to calculate the number of simulations per minute, $spm = N^2/time$, and the efficiency of the code. Efficiency is defined as $Eff = (spm_n/n)/spm_1$, where n is the number of parallelisation threads, spm_n is the number of simulations per minute using n threads, and spm_1 is the number of simulations per minute using 1 thread. Figure 4 (c) shows spm_n and Eff for $N^2 = 128 \times 128$. spm_n shows a linear increase after 2 threads. This shows quantitatively the performance improvement when more parallelisation threads are used. In the case of Eff, it decreases drastically as the number of threads increases, but remains practically constant after 8 threads. We can therefore see that a large improvement in spm_n is achieved by increasing the number of threads, at the cost of only halving the efficiency.

5. Results

In the following, we will show how the software can be used to carry out various analyses, both from the point of view of the physical interpretation of the phenomena observed in the diagram and from the point of view of the performance of the materials. To do this, we will present different results using the thermodynamic approaches described separately in sections 2.1 and 2.2.

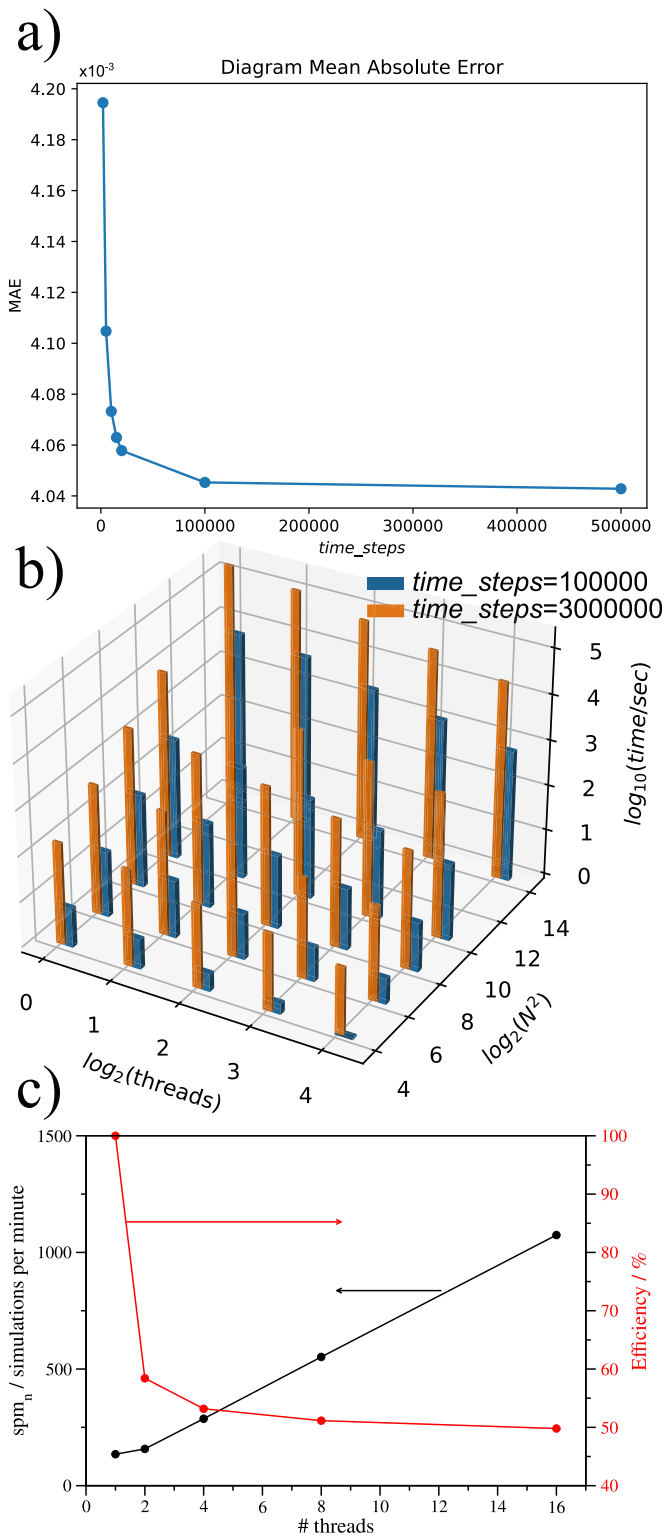

Figure 4. (a) Mean average error in datasets as a function of *time_steps* versus the integral solutions shown in section S1 of the SI. (b) Timerun for *time_steps* = 100000 and *time_steps* = 3000000 as a function of the number of threads and the diagram points. (c) Performance efficiency and simulations per minute by number of threads calculations at $N^2 = 128 \times 128$ with *time_steps* = 100000.

5.1. Langmuir-Frumkin model

We have set $E^0 = 0$V and $E_{cut} = -0.15$V for the calculations performed in this section. The parameter g, defined in equation (6), changes the nature of the interactions between inserted ions. As commented in the Workflow section, the `GalvanostaticMap` class was used to simulate and plot all the galvanostatic diagrams. The `GalvanostaticProfile` class was also implemented to simulate all voltage and concentration profiles. See the Workflow section for details. Figure S3 in the Supplementary information shows three SoC $_{max}$ diagrams for the Langmuir case ($g = 0$), for different particle geometries: planar ($z = 1$), cylindrical ($z = 2$) and spherical

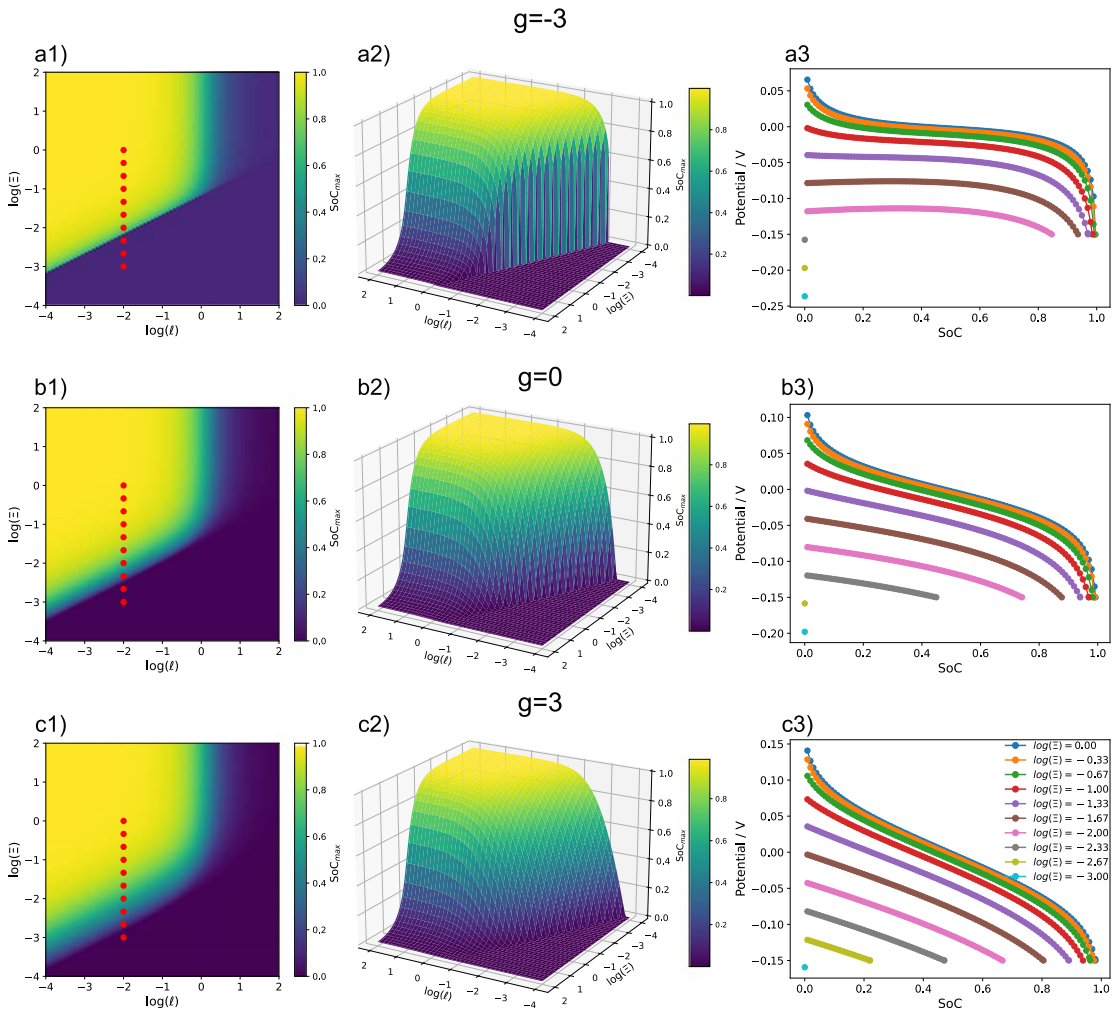

Figure 5. SoC $_{max}$ diagrams using a Frumkin intercalation isotherm for spherical geometry, with different interaction parameters g at log(ℓ)=-2. As the interaction becomes more attractive a stepper jump is observed.

($z = 3$). These are the same results as those reported in the previous work [19], showing the correct functioning of the code to construct these diagrams. In section S3 of the Supplementary information, we include a brief discussion on the impact of the geometric factor and the cut-off potential on the features of the diagram, for a Langmuirian case. In the following, we will focus on showing the effect of Li-Li interactions within the Frumkin approach for the spherical geometry. For more details on the use of the universal diagram to compare material performance, we refer to previous work [20].

A Frumkin isotherm ($g \neq 0$) was used to analyse the effect of ion interactions on the colour maps. This is the first time that such an analysis has been carried out using these diagrams. Figure 5 shows two representative cases: $g = -3$ (figures a1, a2 and a3) and $g = 3$ (figures c1, c2 and c3). The case where $g = 0$ (figures b1, b2 and b3) is used as a reference case. As observed, attractive interactions promote a steeper jump in SoC $_{max}$ compared to $g = 0$ when varying log(Ξ) for a constant log(ℓ). The vertical red lines in the plots illustrate paths where SoC $_{max}$ changes from a large to a small value. The sharpness of the transition in SoC $_{max}$ manifests itself as a colour change, which can be gradual (e.g. $g = 3$) or abrupt (e.g. $g = -3$), as can be seen in the 3D plots. Thus, in general, repulsive interactions produce a smoother and slower change in SoC $_{max}$ compared to attractive interactions.

The previous difference is due to the shape of the voltage profiles for each case, i.e. attractive or repulsive Li-Li interactions. The set of isotherms in figure 5 a3, b3 and c3 helps to understand this. The profiles shown correspond to the red dotted line drawn in each diagram, for a constant log(ℓ) varying log(Ξ). In all cases, it can be observed that lower log(Ξ) promotes a shift of the voltage profile towards the cut-off potential. However, in the case of attractive interactions, the voltage profiles show a potential that remains practically constant over a wide range of SoCs. Thus, for a sufficiently low log(Ξ), there is an abrupt drop in the SoC $_{max}$ between two close log(Ξ) values. This does not happen for repulsive interactions, where the SoC $_{max}$ at the cut-off potential decreases smoothly as log(Ξ) decreases.

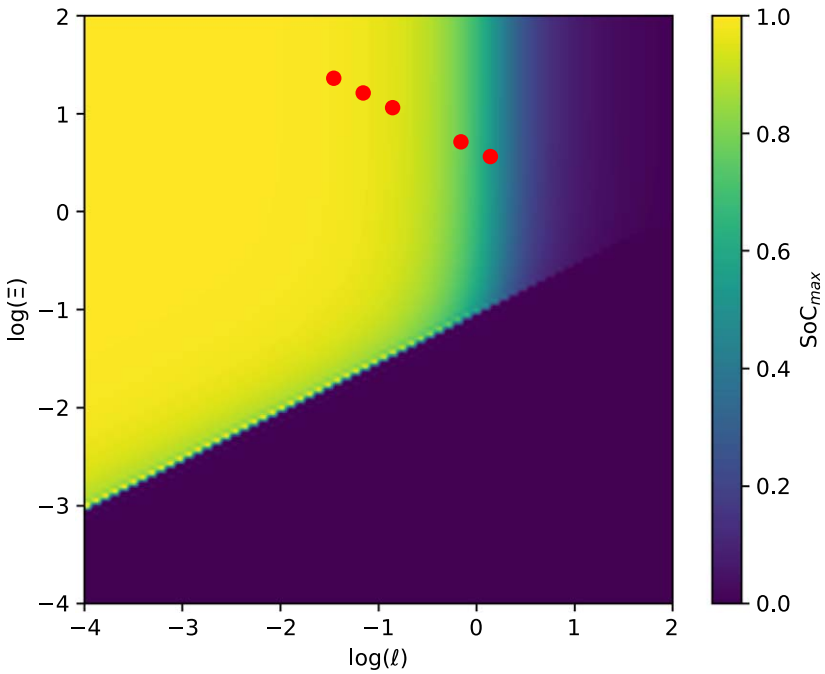

Figure 6. Theoretical diagram using the Frumkin isotherm with $g = -4.2$ [31]. Red dots represent experimental data [40], $C_r = 0.5$C, 1C, 2C, 10C and 20C and $d = 2 \times 10^{-3}$cm, $D = 5 \times 10^{-9}$cm s^-1^2 and $k^0 = 1 \times 10^{-5}$cm s^-1.

Another utility of the software is the generation of concentration profiles. Figure S5a) shows a Langmuir voltage profile generated with the `GalvanostaticProfile` class, for a point of the diagram corresponding to log(Ξ)=-1 and log(ℓ)=0. Different SoC, marked with coloured points in figure S5a), were selected to calculate the concentration profiles shown in figure S5b), using the `concentration_dataframe` and `concentration_plot` methods. The coordinate r_{norm} describes the distance from the centre of the particle, $r_{norm} = 0$, to the surface, $r_{norm} = 1$, and x is the degree of lithium-ion occupation. As we reach the surface, x increases exponentially, with the average occupancy becoming progressively higher as the SoC value increases, as expected.

Focusing on a specific case, it is possible to reproduce a material diagram if the value of g is known. For example, in the case of LCO, Levi *et al* found the value of the interaction parameter to be $g = -4.2$ [31]. A diagram constructed with this Frumkin interaction parameter is shown in figure 6. As expected, the resulting plot presents the characteristics described above for attractive interactions. Wang *et al* worked with LCO single particle electrodes [40] and they measured the rate capability for many C_r. The C-rates of the experimental setup were $C_r = 0.5$C, 1C, 2C, 10C and 20C. The diffusion length, diffusion coefficient and kinetic constant used were those reported: $d = 2 \times 10^{-3}$cm, $D = 5 \times 10^{-9}$cm s^-1^2 and $k^0 = 1 \times 10^{-5}$cm s^-1. The plots of these experimental points are marked in figure 6 with red dots. Good agreement is observed between the diagram SoC $_{max}$ and the reported values [40].

5.2. Interpolation of experimental data to get intercalation isotherms

Intercalation isotherms represent the amount of Li ions intercalated into a certain material as a function of the electrode potential under equilibrium conditions. The use of the experimental equilibrium intercalation isotherms involves a new feature: the equilibrium potential at different loadings of the material $E^0(x_d)$ must be obtained from the interpolation of experimental data points. As commented, the data interpolation is performed with the `CubicSpline` function of the `scipy.interpolate` package. In the experimental context, we considered occupation/voltage profile curves that overlap under sufficiently small C-rates. In this section, spherical particles were considered in all cases.

A diagram constructed using the equilibrium intercalation isotherm of LMO is shown in figure 7(a). The experimental data points were taken from Bak *et al* [41], where the intercalation isotherm for $C_r = 1$C was taken as the equilibrium isotherm as it overlaps with that for $C_r = 0.5$C. The cut-off potential was set to $E_{cut} = 3.8$V versus Li/Li^+. An important detail to note is that for the construction of the diagram, it is not necessary to provide realistic values for k^0, D, d and C_r, since Ξ and ℓ are invariant to the possible combinations of these parameters. Instead, for example, arbitrary values of k^0 and D can be provided to calculate d and C_r with equations (8) and (9). In the present software default values of k^0 and D are provided. Returning to the diagram

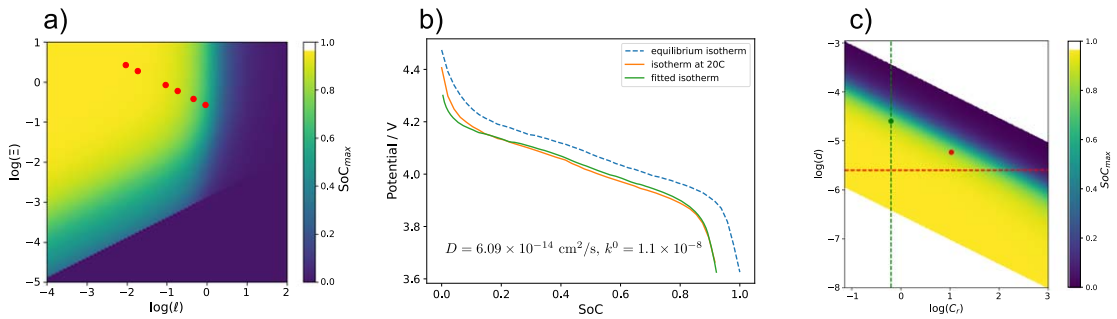

Figure 7. (a) SoC $_{max}$ LMO diagram, $E_{cut} = 3.8$V versus Li/Li$^+$. (b) Isotherm fitting at 20C, $D = 6.09 \times 10^{-14}$ cm^2/s, $k^0 = 1.1 \times 10^{-8}$. c) Diagram in the log(d) versus log(C_r) space, d in cm.

in figure 7(a), differences relative to the Langmuir model are observed. This shows the influence of the LMO thermodynamics. In particular, a slow decrease in SoC $_{max}$ is observed as log(Ξ) decreases. This is reminiscent of the effect of repulsive interactions in the Frumkin model described in the previous section. However, as shown in previous work [42–44], the thermodynamics of LMO cannot be simplified to a single type of interaction.

The work of Back *et al* [41] provides the C-rate and the particle size d of LMO particles, however, no information regarding k^0 and D is given. So, the constructed diagram can be employed for obtaining these parameters by using our fitting tool `ProfileFitting`.

Figure 7(b) shows the results of the fitting routine applied to an objective LMO intercalation isotherm obtained at $C_r = 20$ C , with $d = 2.5 \times 10^{-6}$ cm, from the work of Bak *et al* [41]. The values obtained were $D = 6.09 \times 10^{-14}$ cm^2/s and $k^0 = 1.1 \times 10^{-8}$ cm/s. We can compare these predictions with experimental literature data, using Li$^+$ as a tracer with neutron radiography [45] and secondary ion mass spectrometry [46]. These authors found values of the Li$^+$ diffusion coefficient of the order of 10^{-14} cm^2 s^{-1}. This result also agrees with those obtained by electrochemical means [47]. With the fitted parameters and d, we can calculate the position of the experiments of Bak *et al* [41] in the diagram for $C_r = 1$C, 2C, 10C, 20C, 50C and 100C, using the equations (8) and (9). These are shown as red spots in figure 7(a). As can be seen, the only data point below a SoC $_{max}$ of 0.8 corresponds to 100C, in agreement with the experimental observations.

We can also show another possible strategy for visualising the data for this experimental system using the theoretical diagram. Since k^0 and D are usually treated as constant values, and since d and C_r are known for an experimental setup, one can use equations (8) and (9) to transform the dimensionless plot log(Ξ) versus log(ℓ) into a dimensional one, for example log(d) versus log(C_r). This is what we have done for the data of Bak *et al* in figure 7(c), for a range of values of experimental applicability. The red horizontal line shows the changes in the C-rate in Bak's experimental setup with $d = 2.5 \times 10^{-6}$ cm. This kind of representation provides information quickly and visually, allowing different predictions to be made, such as rapid charging conditions, the charging rate required to achieve a given SoC, and knowing the limitations and materials potential. For example, according to this plot, the largest particle size that may be synthesised achieving a SoC = 0.8 at a C-rate of 10 C is $d = 5.77 \times 10^{-6}$ cm. This point is marked with a red dot in figure 7(c). Similarly, the maximum C-rate that may be applied to obtain a SoC around 0.8 for a particle size of $d = 2.5 \times 10^{-5}$ cm (vertical l green line) is $C_r = 0.63$C. The diagram shows this predicted point with a green dot.

The case of graphite is illustrative in observing how the thermodynamics of a material moulds the shape of the diagram. Graphite is a material that undergoes several phase transitions as lithium intercalation proceeds. These transitions occur at fixed potentials, giving each a plateau in the voltage profile. Levi *et al* obtained the value of g for some of these stages by fitting the peaks of the cyclic voltammogram [48]. In the present approach, instead of taking a stepwise function with different values of g, the experimental isotherm and the interpolation method take into account the whole thermodynamics of the system. Figure 8 (a) shows different charging curves simulated for graphite at a constant value of log(ℓ)=-2 by varying log(Ξ), using experimental data for single particle graphite electrodes at quasi-equilimbirum conditions taken from the work of Tao *et al* [49]. Figure 8 (b) shows the 2D colour map obtained using the data from Tao *et al.*In this case $E_{cut} = 0.01$V versus Li/Li$^+$. Figures 8(b) and (c) show the resulting 2D colour map and the three-dimensional representation of SoC $_{max}$. The intercalation phases are observed as colour steps in the diagram and the 3D plot. Also, at lower log(ℓ), the shape of the 3D plot resembles that of the equilibrium isotherm.

To compare with the LCO diagram in figure 6, simulated with the Frumkin model, we have used a LCO equilibrium curve to generate a diagram. This is shown in figure 9. The equilibrium isotherm was taken from the work of Wang *et al* [40], using a cut-off potential of 3.75V versus Li/Li$^+$. Considering that the equilibrium potential is 3.9V versus Li/Li$^+$, we get a potential difference $E - E_{cut} = -150$ mV relative to equilibrium. The

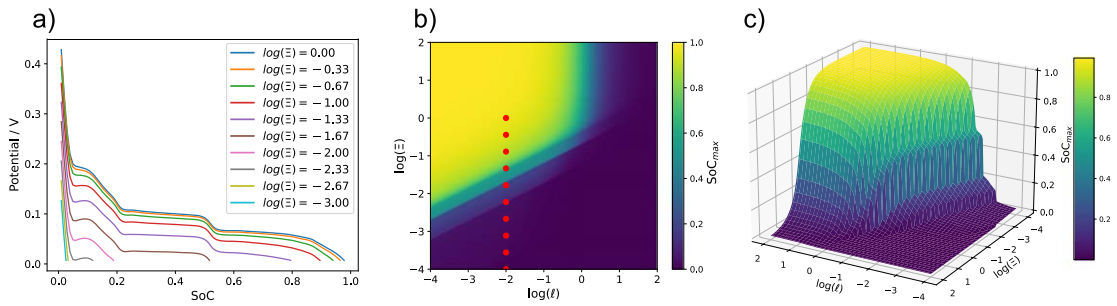

Figure 8. (a) Simulated charging curves for graphite, for varying log(Ξ) at log (ℓ)=-2, E_{cut} = 0.01V. (b) SoC $_{max}$ graphite diagram, E_{cut} = 0.01V. (c) Three-dimensional diagram protection, for low ℓ the isotherm plateau are observed.

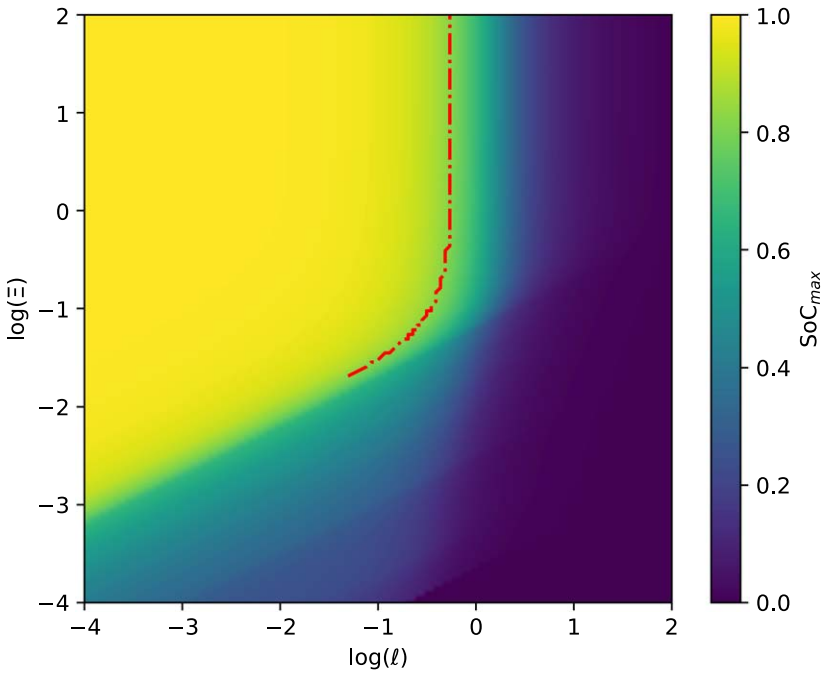

Figure 9. SoC $_{max}$ colour diagram constructed using experimental discharge isotherm at 0.5C from [40]. The red dotted line denotes contour where SoC$_{max}$ ~ 0.8 in the LCO Frumkin diagram, figure 6.

similarity with the Frumkin plot of figure 6 is striking. Above the straight line with slope 1/2, in the log(Ξ) − log(ℓ) region given by $R = (-3, \infty) \times (-4, \infty)$ the colour plots are similar. The red dotted line in figure 9, denotes the contour where SoC$_{max}$ ~ 0.8 in the LCO Langmuir diagram of figure 6. It can be seen that both regions overlap, showing a good agreement between the two models. This is a remarkable feature of the theoretical model. However, below the region described, the realistic diagram shows a smooth decrease in the state of charge as log(Ξ) decreases, whereas in the Frumkin model, the SoC drops sharply to zero. This highlights the need to include the thermodynamics of the material, if necessary, to obtain an accurate representation.

6. Conclusions and perspectives

We present a powerful software tool designed to create galvanostatic maps, offering critical insights into the charging characteristics of lithium-ion battery single-particle materials and related systems. The diagram built by the program allows the user to make different analyses and predictions regarding the maximum capacity reachable by a single-particle material, given the particle size, the diffusion coefficient, the charge transfer rate and the C-rate (See figure 1 for an illustration). Thus, this software allows making various predictions that are very useful for optimizing active materials under fast charging conditions. The inclusion of the insertion isotherm in the construction of the maps represents an improvement to previous modeling. The results showed that the inclusion of thermodynamics of materials modifies the features of the galvanostatic diagrams, adopting

the shape of the equilibrium isotherm, although the zone above log(Ξ)=0 remains essentially unchanged. This is an important result that needs further investigation. An interesting feature of the dimensionless diagram is that it is not necessary to know the kinetic parameters to build them. However, it can be applied as a fitting tool to obtain the kinetic parameters k^0 and D, given the C-rate and the particle size of an specific experimental setup. Also, these dimensionless diagrams, specific for each material, allows constructing alternative diagrams by converting it to a dimension one, if the diffusion coefficient and the charge transfer constant are considered to remain constant. This log C-rate versus log particle size representation allows making fast-charging predictions, for example, and delimiting the area where it is safest to work with the model, given the limitation in not considering the dynamics of the electrolyte.

The software seamlessly integrates a user-friendly Python API with a high-performance computational core developed in parallel C++, ensuring both speed and efficiency. Our tool demonstrates remarkable versatility, capable of generating diagrams using various thermodynamic models. This flexibility enables comprehensive predictions and analyses, evaluating the performance of different intercalation materials. Additionally, the software supports the calculation of chronopotentiometric and concentration profiles, providing essential data to interpret the underlying physics of the map features. A notable feature of the software is its fitting procedure, which allows the use of these diagrams to match out-of-equilibrium potential profiles, thereby extracting valuable kinetic information from intercalation systems. This advanced functionality makes our tool indispensable for researchers aiming to diagnose and optimize the performance of alkali-ion battery materials. Currently, there is no software like the one presented here to create this type of diagram, with the versatility of being able to switch between different thermodynamic models, and thus perform different analyses of the intercalation materials. Thus, using the suitable parameters, the present model may also be used to optimize the behavior of Na-ion and K-ion [50, 51].

We plan to expand the current software by integrating refined models that will encompass additional phenomena within the existing theoretical framework. These advancements will include electrolyte dynamics to provide a more comprehensive understanding of its influence on battery performance. Additionally, artificial intelligence and machine learning techniques [2] may help to predict galvanostatic diagrams more effectively, allowing for the inclusion of various factors such as the impact of additives and binders on rate performance. Furthermore, improvements in the diffusion model are expected, with the potential application of Physics-Informed Neural Networks (PINN) to enhance the accuracy of simulations.

Acknowledgments

The authors acknowledge the financial support from MAHIR Project, the First Argentine-Israel Scientific Research Program (MINCyT-MOST). E.M. Gavilán-Arriazu acknowledges grant FONCYT PICT-2020-SERIEA-00707. F. F. acknowledges support from a Ph.D. fellowship from CONICET. E.P.M. Leiva acknowledges grants PIP CONICET 1220200101189CO, PUE/2017 CONICET, FONCYT 2020-SERIEA-03689 and SECyT of the Universidad Nacional de Córdoba. Support by CCAD-UNC and GPGPU Computing Group, Y-TEC and an IPAC grant from SNCAD-MinCyT, Argentina, are also gratefully acknowledged. Y. Ein-Eli acknowledges the financial and infrastructure support from the Israeli National Institute for Energy Storage (I-NIES), and Grand Technion Energy Program. We would like to thank Dr. Nicolas Wolovick for his helpful comments and Dr. Fernanda Stragliotto and Dr. Esteban Euti for administrative support.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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